

Computational model

Our model consists of three neuronal populations (C_1 , C_2 , and O), artificially separated for clarity rather than representing strictly segregated neural populations. To generate activity from the animal's simulated trajectories, we employ a linear-non-linear Poisson approach: simulating firing rate dynamics via a continuous attractor neural network (CANN) and generating discrete spikes from a Poisson distribution parameterized by the computed rates.

$$\begin{aligned}\tau \frac{dU_i^{(C_1)}}{dt} &= -U_i^{(C_1)} + \exp\left(-\frac{d(\theta(t), \theta_i)^2}{2\sigma_\theta^2}\right) \left(\sum_j W_{ij}^{\text{rec}} f(U_j^{C_1}) + I_{vis}\right), \\ \tau \frac{dU_i^{(C_2)}}{dt} &= -U_i^{(C_2)} + \alpha_i(t) \exp\left(-\frac{d(\theta(t), \theta_i)^2}{2\sigma_\theta^2}\right) \left(\sum_j W_{ij}^{C_1 \rightarrow C_2} f(U_j^{C_1})\right), \\ \tau \frac{dU_i^{(O)}}{dt} &= -U_i^{(O)} + \sum_j W_{ij}^{C_2 \rightarrow O} f(U_j^{C_2}) - I_0,\end{aligned}\quad (1)$$

Network Dynamics

For any neuron i in the cell population p , its internal state variable $U_i^{(p)}(t)$ integrates synaptic inputs firing rate dynamics:

$$\tau \frac{dU_i^{(p)}}{dt} = -U_i^{(p)} + I_{in,i}^{(p)}(t), \quad (2)$$

where τ is the membrane time constant and $I_{in,i}(t)$ is the total net input current to the cell. The firing rate $r_i^{(p)}(t)$ is modeled as a nonlinear function of the internal state variable:

$$r_i^{(p)}(t) = g^{(p)} \frac{[U_i^{(p)}(t) - T^{(p)}]_+^2}{1 + k \sum_m [U_m^{(p)}(t) - T^{(p)}]_+^2}, \quad (3)$$

where $g^{(p)} > 0$ is a gain factor, $T^{(p)} > 0$ a linear population threshold, $[\cdot]_+$ rectifies the content taking the maximum value between it and zero and k determines the global inhibition strength that stabilizes activity and prevents runaway excitation. Spike counts for a given time bin of length Δt are then obtained through a pseudo-random Poisson process:

$$P(k \text{ spikes}) = \frac{(r_i(t)\Delta t)^k e^{-r_i(t)\Delta t}}{k!}$$

Topology and Connectivity

Neurons in each population are assigned a preferred spatial phase $\Phi_i = (\Phi_i^x, \Phi_i^y)$ on a rhombic domain, corresponding to a hexagonal lattice of firing fields in

physical space. Conjunctive cells are additionally assigned a preferred head direction θ_i . Synaptic connections depend on the phase distance on a twisted toroidal manifold, $\|\cdot\|_{Torus}$, which accounts for the periodic boundaries of the grid lattice.

Feedforward weights from a conjunctive population to another grid cell population ($C_1 \rightarrow C_2$ as well as $C_2 \rightarrow O$) take the form

Primary Conjunctive Layer (C_1)

The first layer comprises primary conjunctive grid-by-head-direction cells. These cells integrate self-motion and spatial inputs. Their input current $I_{in,i}^{C_1}$ is driven by bottom-up spatial visual inputs (I_{vis}), local head direction tuning (I_{hd}), and internal recurrent collateral connections (W^{rec}):

$$I_{in,i}^{C_1}(t) = a_i \left(I_{vis,i}(t) + \sum_j W_{ij}^{rec} r_j^{C_1}(t) \right), \quad \text{with} \quad a_i = \exp \left(-\frac{d(\theta(t), \theta_i^{C_1})^2}{2\sigma_{HD}^2} \right) \quad (4)$$

The recurrent weights W_{ij}^{rec} between a presynaptic cell j and postsynaptic cell i follow a Gaussian profile centered on a shifted phase:

$$W_{ij}^{rec} = \exp \left(-\frac{\|\Phi_j - \Phi_i - \Delta_{HD} \hat{e}(\theta_j)\|_{Torus}^2}{2\sigma^2} \right)$$

where σ controls the size of the synaptic footprint, and Δ_{HD} is an asymmetric shift along the presynaptic preferred direction $\hat{e}(\theta_j)$ to drive network translation during movement. Spatial maps in this layer do not shift upward on inclined surfaces.

Intermediate Conjunctive Layer (C_2)

The second layer consists of intermediate conjunctive cells driven by feedforward projections from C_1 and modulated by vestibular signals. The net input current is:

$$I_{in,i}^{C_2}(t) = a_i(1 + b_i \sin(\alpha)) \sum_j W_{ij}^{1 \rightarrow 2} r_j^{C_1}(t) \quad (5)$$

The feedforward weights $W_{ij}^{1 \rightarrow 2}$ share the identical shifted Gaussian topology as the recurrent weights, projecting a predicted spatial phase. The vestibular gain $A_{vest,i}(t)$ incorporates slope-dependent modulation:

$$A_{vest,i}(t) = 1 + \beta_{vest} \sin(\alpha) \exp \left(-\frac{(\theta_S - \theta_i)^2}{2\sigma_S^2} \right)$$

where β_{vest} controls the strength of the modulation associated with slope angle α in the inclination direction θ_S , and σ_S dictates the directional tuning. While this gain does not affect path integration in the primary layer, it enhances the feedforward contribution of cells with an uphill-pointing preferred direction.

Omnidirectional Grid Layer (O)

The final layer comprises omnidirectional grid cells. These cells lack intrinsic directional preference and receive convergent feedforward input from C_2 :

$$I_{in,i}^O(t) = \sum_j W_{ij}^{2 \rightarrow o} r_j^{C_2}(t)$$

The weights $W_{ij}^{2 \rightarrow o}$ map the directionally tuned predictive activity back into a purely spatial representation. Because C_2 activity is selectively amplified by the vestibular gain, the resulting center-of-mass in the O layer shifts upward on inclined surfaces, reproducing the experimentally observed non-directional shift.